

COMPLETE VISUALIZATION CHEAT SHEET - TITANIC DATASET

Quick Reference for All Plot Types

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MATPLOTLIB BASICS

Figure Creation

```
# Single plot
fig, ax = plt.subplots()
fig, ax = plt.subplots(figsize=(10, 6))

# Multiple subplots
fig, axes = plt.subplots(2, 2)
fig, axes = plt.subplots(2, 3, figsize=(15, 10))

# Access subplots
ax = axes[0, 0] # Row 0, Col 0
ax = axes.flatten()[0] # After flattening
```

Common Customizations

```
ax.set_title('Title', fontsize=16, fontweight='bold')
ax.set_xlabel('X Label', fontsize=12)
ax.set_ylabel('Y Label', fontsize=12)
ax.grid(True, alpha=0.3)
ax.legend(loc='best')
ax.set_xlim(0, 100)
ax.set_ylim(0, 1)
plt.tight_layout()
plt.savefig('plot.png', dpi=300, bbox_inches='tight')
```

UNIVARIATE PLOTS

1. HISTOGRAM

What: Frequency distribution of continuous data

When: See distribution shape, central tendency, spread

Data: One numerical (continuous)

```
# Matplotlib
plt.hist(df['Age'], bins=30, color='skyblue', edgecolor='black')

# Seaborn - Basic
sns.histplot(data=df, x='Age')
sns.histplot(data=df, x='Age', bins=30)
sns.histplot(data=df, x='Age', kde=True) # Add density curve

# By category
sns.histplot(data=df, x='Age', hue='Survived')
sns.histplot(data=df, x='Age', hue='Survived', multiple='stack')
sns.histplot(data=df, x='Age', hue='Survived', multiple='dodge')
sns.histplot(data=df, x='Age', hue='Survived', multiple='layer', alpha=0.5)

# Titanic Example
sns.histplot(data=df, x='Age', bins=30, kde=True, color='steelblue')
# Shows: Most passengers 20-40 years old
```

Key Parameters:

- **bins:** Number of bins (default: auto)
 - **kde:** Add kernel density estimate
 - **hue:** Color by category
 - **multiple:** How to handle multiple distributions ('layer', 'stack', 'dodge')
 - **stat:** 'count', 'density', 'probability'
 - **alpha:** Transparency (0-1)
-

2. KDE PLOT

What: Smooth probability density curve

When: Alternative to histogram, smooth representation

Data: One numerical (continuous)

```
# Basic
sns.kdeplot(data=df, x='Age')
sns.kdeplot(data=df, x='Age', fill=True)

# By category
sns.kdeplot(data=df, x='Age', hue='Survived', fill=True, alpha=0.5)
sns.kdeplot(data=df, x='Age', hue='Survived', common_norm=False)

# Bandwidth (smoothness)
sns.kdeplot(data=df, x='Age', bw_adjust=0.5) # Less smooth
sns.kdeplot(data=df, x='Age', bw_adjust=2) # More smooth
```

```
# Cumulative
sns.kdeplot(data=df, x='Age', cumulative=True)

# Titanic Example
sns.kdeplot(data=df, x='Age', hue='Survived', fill=True, alpha=0.5)
# Shows: Age distribution overlap between survivors/non-survivors
```

Key Parameters:

- `fill`: Fill area under curve
 - `bw_adjust`: Bandwidth multiplier (smoothness)
 - `cumulative`: Show CDF instead of PDF
 - `common_norm`: Normalize distributions separately
-

3. BOX PLOT

What: Five-number summary + outliers

When: See quartiles, median, outliers

Data: One or more numerical variables

```
# Basic
sns.boxplot(data=df, y='Age')

# By category
sns.boxplot(data=df, x='Pclass', y='Age')
sns.boxplot(data=df, x='Pclass', y='Age', hue='Survived')

# Horizontal
sns.boxplot(data=df, x='Age', y='Pclass', orient='h')

# Customization
sns.boxplot(data=df, x='Pclass', y='Fare', showfliers=False) # Hide outliers
sns.boxplot(data=df, x='Pclass', y='Age', showmeans=True)

# Titanic Example
sns.boxplot(data=df, x='Pclass', y='Fare')
# Shows: 1st class paid much more, many fare outliers
```

Key Parameters:

- `showfliers`: Show/hide outliers
- `showmeans`: Show mean marker
- `width`: Box width
- `palette`: Color scheme

Box Components:

- Box: Q1 to Q3 (IQR)
- Line in box: Median
- Whiskers: $1.5 \times \text{IQR}$

- Points: Outliers
-

4. VIOLIN PLOT

What: Box plot + KDE

When: More detail than box plot

Data: Numerical by categories

```
# Basic
sns.violinplot(data=df, x='Pclass', y='Age')

# By category
sns.violinplot(data=df, x='Pclass', y='Age', hue='Survived')
sns.violinplot(data=df, x='Pclass', y='Age', hue='Survived', split=True)

# Inner representation
sns.violinplot(data=df, x='Pclass', y='Age', inner='quartile')
sns.violinplot(data=df, x='Pclass', y='Age', inner='box')
sns.violinplot(data=df, x='Pclass', y='Age', inner='point')
sns.violinplot(data=df, x='Pclass', y='Age', inner=None)

# Scale
sns.violinplot(data=df, x='Pclass', y='Age', scale='width')      # Same width
sns.violinplot(data=df, x='Pclass', y='Age', scale='area')      # Same area
sns.violinplot(data=df, x='Pclass', y='Age', scale='count')     # By count

# Titanic Example
sns.violinplot(data=df, x='Pclass', y='Age', hue='Survived', split=True)
# Shows: Age distribution shape for each class and survival
```

Key Parameters:

- **split:** Split violins for hue
 - **inner:** What to show inside ('quartile', 'box', 'point', None)
 - **scale:** Violin width method
 - **cut:** Extend beyond data (default: 2)
-

5. COUNT PLOT

What: Bar chart of categorical frequencies

When: Count occurrences of categories

Data: One categorical

```
# Basic
sns.countplot(data=df, x='Pclass')
sns.countplot(data=df, y='Embarked') # Horizontal

# By category
sns.countplot(data=df, x='Pclass', hue='Survived')

# Order
sns.countplot(data=df, x='Embarked', order=['S', 'C', 'Q'])
```

```
# Titanic Examples
sns.countplot(data=df, x='Survived')
# Shows: 62% died (0), 38% survived (1)

sns.countplot(data=df, x='Pclass', hue='Survived')
# Shows: Most deaths in 3rd class
```

Key Parameters:

- `order`: Specify category order
 - `hue`: Split by another category
 - `palette`: Color scheme
-

BIVARIATE PLOTS

6. SCATTER PLOT

What: Points showing relationship

When: See correlation, patterns

Data: Numerical × Numerical

```
# Basic
sns.scatterplot(data=df, x='Age', y='Fare')

# Color by third variable
sns.scatterplot(data=df, x='Age', y='Fare', hue='Survived')

# Size by third variable
sns.scatterplot(data=df, x='Age', y='Fare', size='Pclass')

# Style by third variable
sns.scatterplot(data=df, x='Age', y='Fare', style='Sex')

# Combine
sns.scatterplot(data=df, x='Age', y='Fare',
                hue='Survived', size='Pclass', style='Sex', alpha=0.6)

# Titanic Example
sns.scatterplot(data=df, x='Age', y='Fare', hue='Survived', alpha=0.6)
# Shows: No strong correlation, survivors across all ages/fares
```

Key Parameters:

- `hue`: Color by variable
 - `size`: Size by variable
 - `style`: Marker style by variable
 - `alpha`: Transparency
 - `s`: Fixed marker size
-

7. BAR PLOT

What: Aggregated values by category

When: Compare means/medians across groups

Data: Categorical × Numerical

```
# Basic (shows mean)
sns.barplot(data=df, x='Pclass', y='Fare')

# Different estimator
sns.barplot(data=df, x='Pclass', y='Fare', estimator='median')
sns.barplot(data=df, x='Pclass', y='Age', estimator=np.std)

# By category
sns.barplot(data=df, x='Pclass', y='Fare', hue='Survived')

# Error bars
sns.barplot(data=df, x='Pclass', y='Fare', ci=95)      # 95% CI
sns.barplot(data=df, x='Pclass', y='Fare', ci='sd')   # Std dev
sns.barplot(data=df, x='Pclass', y='Fare', ci=None)   # No error bars

# Titanic Examples
sns.barplot(data=df, x='Pclass', y='Survived')
# Shows: Survival rate by class (1st > 2nd > 3rd)

sns.barplot(data=df, x='Sex', y='Survived')
# Shows: Females survived at much higher rate
```

Key Parameters:

- **estimator:** Aggregation function (default: mean)
 - **ci:** Confidence interval (95, 'sd', None)
 - **hue:** Split by category
 - **order:** Specify category order
-

8. REGRESSION PLOT

What: Scatter + regression line

When: See trend and correlation

Data: Numerical × Numerical

```
# Linear regression
sns.regplot(data=df, x='Age', y='Fare')

# Polynomial
sns.regplot(data=df, x='Age', y='Fare', order=2) # Quadratic

# LOWESS (smooth)
sns.regplot(data=df, x='Age', y='Fare', lowess=True)

# Customize
sns.regplot(data=df, x='Age', y='Fare',
            scatter_kws={'alpha':0.3, 's':30},
            line_kws={'color':'red', 'linewidth':2})

# Logistic regression (binary outcome)
```

```
sns.regplot(data=df, x='Age', y='Survived', logistic=True)

# With faceting (lmplot)
sns.lmplot(data=df, x='Age', y='Fare', col='Pclass')
sns.lmplot(data=df, x='Age', y='Fare', hue='Survived')

# Titanic Example
sns.regplot(data=df, x='Fare', y='Survived', logistic=True, ci=None)
# Shows: Higher fare slightly increases survival probability
```

Key Parameters:

- **order**: Polynomial degree
- **logistic**: Logistic regression for binary
- **lowess**: Local regression (smooth)
- **robust**: Robust regression (outlier-resistant)
- **ci**: Confidence interval

9. JOINT PLOT

What: Bivariate + marginal distributions

When: See relationship AND individual distributions

Data: Numerical × Numerical

```
# Basic
sns.jointplot(data=df, x='Age', y='Fare')

# Different types
sns.jointplot(data=df, x='Age', y='Fare', kind='scatter')
sns.jointplot(data=df, x='Age', y='Fare', kind='hex')
sns.jointplot(data=df, x='Age', y='Fare', kind='kde')
sns.jointplot(data=df, x='Age', y='Fare', kind='reg')

# With hue
sns.jointplot(data=df, x='Age', y='Fare', hue='Survived')

# Titanic Example
sns.jointplot(data=df, x='Age', y='Fare', kind='scatter', alpha=0.5)
# Shows: Central scatter + age distribution + fare distribution
```

Key Parameters:

- **kind**: 'scatter', 'hex', 'kde', 'reg', 'resid'
- **hue**: Color by category
- **height**: Figure size

MULTIVARIATE PLOTS

10. PAIR PLOT

What: All pairwise scatter plots

When: Explore all relationships

Data: Multiple numerical variables

```
# Basic
sns.pairplot(data=df)

# Select columns
sns.pairplot(data=df[['Age', 'Fare', 'SibSp', 'Parch', 'Survived']])

# Color by category
sns.pairplot(data=df, hue='Survived')
sns.pairplot(data=df, hue='Pclass', palette='Set1')

# Diagonal type
sns.pairplot(data=df, hue='Survived', diag_kind='kde')
sns.pairplot(data=df, kind='reg') # All regression plots

# Corner (lower triangle only)
sns.pairplot(data=df, corner=True)

# Titanic Example
sns.pairplot(data=df[['Age', 'Fare', 'Survived']], hue='Survived')
# Shows: All relationships between Age, Fare, colored by survival
```

Key Parameters:

- **hue:** Color by category
 - **diag_kind:** 'hist' or 'kde'
 - **kind:** 'scatter' or 'reg'
 - **corner:** Show only lower triangle
 - **height:** Size of each subplot
-

11. HEAT MAP

What: Color-coded matrix

When: Show correlation, pivot tables

Data: Matrix of numbers

```
# Correlation matrix
corr = df[['Age', 'Fare', 'SibSp', 'Parch']].corr()
sns.heatmap(corr)

# With annotations
sns.heatmap(corr, annot=True, fmt='.2f')

# Color map
sns.heatmap(corr, cmap='coolwarm', center=0)

# Square cells
```

```

sns.heatmap(corr, square=True)

# Mask upper triangle
mask = np.triu(np.ones_like(corr, dtype=bool))
sns.heatmap(corr, mask=mask, annot=True, cmap='coolwarm')

# Pivot table
pivot = df.pivot_table(values='Survived', index='Pclass', columns='Sex')
sns.heatmap(pivot, annot=True, fmt='.2f', cmap='RdYlGn')

# Titanic Example
cols = ['Survived', 'Pclass', 'Age', 'Fare', 'SibSp', 'Parch']
sns.heatmap(df[cols].corr(), annot=True, fmt='.2f', cmap='coolwarm')
# Shows: Correlation between all variables

```

Key Parameters:

- **annot:** Show values
 - **fmt:** Number format
 - **cmap:** Color map
 - **center:** Center value for diverging colormap
 - **square:** Square cells
 - **linewidths:** Grid lines
 - **mask:** Hide cells
 - **vmin, vmax:** Value range
-

12. FACET GRID

What: Grid of subplots by categories

When: Compare across categories

Data: Any with categorical variable

```

# Basic
g = sns.FacetGrid(df, col='Pclass')
g.map(sns.histplot, 'Age')

# Row and column
g = sns.FacetGrid(df, row='Pclass', col='Sex')
g.map(sns.histplot, 'Age')

# With hue
g = sns.FacetGrid(df, col='Pclass', hue='Survived')
g.map(sns.scatterplot, 'Age', 'Fare', alpha=0.6)
g.add_legend()

# Customization
g = sns.FacetGrid(df, col='Pclass', height=4, aspect=1.2)
g.map(sns.histplot, 'Age', bins=20)
g.set_titles("Class {col_name}")
g.set_axis_labels("Age", "Count")

# Titanic Example

```

```
g = sns.FacetGrid(df, col='Pclass', row='Sex', hue='Survived', height=3)
g.map(sns.scatterplot, 'Age', 'Fare', alpha=0.5)
g.add_legend()
# Shows: 6 subplots (3 classes × 2 sexes) with survival colored
```

Key Parameters:

- `col`: Column facet variable
 - `row`: Row facet variable
 - `hue`: Color variable
 - `height`: Height of each subplot
 - `aspect`: Width/height ratio
 - `sharex, sharey`: Share axes
-

13. CAT PLOT

What: Categorical plot with faceting

When: Categorical analysis across groups

Data: Categorical × Numerical

```
# Different kinds
sns.catplot(data=df, x='Pclass', y='Age', kind='box')
sns.catplot(data=df, x='Pclass', y='Age', kind='violin')
sns.catplot(data=df, x='Pclass', y='Age', kind='strip')
sns.catplot(data=df, x='Pclass', y='Age', kind='swarm')
sns.catplot(data=df, x='Pclass', y='Age', kind='bar')
sns.catplot(data=df, x='Pclass', y='Age', kind='point')

# With faceting
sns.catplot(data=df, x='Pclass', y='Age', kind='box', col='Sex')
sns.catplot(data=df, x='Pclass', y='Age', kind='violin',
            col='Sex', row='Survived')

# Titanic Example
sns.catplot(data=df, x='Pclass', y='Survived', kind='bar', col='Sex')
# Shows: Survival rate by class, separately for males and females
```

Key Parameters:

- `kind`: Plot type
 - `col, row`: Faceting variables
 - `hue`: Color variable
 - `height, aspect`: Size
-

SEABORN ATTRIBUTES & METHODS

Most Common Attributes

```
# Data
data=df                # DataFrame
x='Age'                # Column for x-axis
y='Fare'               # Column for y-axis

# Aesthetics
hue='Survived'         # Color by variable
size='Pclass'          # Size by variable
style='Sex'            # Marker style by variable
palette='Set2'         # Color scheme
color='steelblue'      # Single color
alpha=0.6              # Transparency (0-1)

# Bins and bandwidth
bins=30                # Number of bins (histogram)
binwidth=5             # Width of bins
bw_adjust=1.5          # Bandwidth adjustment (KDE)

# Statistical
estimator='mean'       # Aggregation function
ci=95                  # Confidence interval
ci='sd'                # Standard deviation
ci=None                # No error bars

# Categorical order
order=['S', 'C', 'Q'] # Specify category order
hue_order=[1, 0]      # Hue category order

# Markers
s=50                   # Marker size
marker='o'             # Marker shape
markers=['o', 's']     # Different markers for hue
edgecolor='black'      # Edge color
linewidth=0.5          # Line/edge width

# Faceting
col='Pclass'           # Column facet
row='Sex'              # Row facet
height=4               # Subplot height
aspect=1.2             # Width/height ratio

# Plot-specific
kde=True               # Add KDE (histogram)
fill=True              # Fill area (KDE)
multiple='layer'       # How to show multiple (histogram)
split=True             # Split violins
inner='quartile'       # Violin interior
showfliers=False       # Hide outliers (box)
logistic=True          # Logistic regression
order=2                # Polynomial order
```

Style and Theme Methods

```
# Set style
sns.set_style('darkgrid')
sns.set_style('whitegrid')
sns.set_style('dark')
sns.set_style('white')
```

```

sns.set_style('ticks')

# Set context (scale)
sns.set_context('paper')
sns.set_context('notebook')
sns.set_context('talk')
sns.set_context('poster')

# Set palette
sns.set_palette('deep')
sns.set_palette('Set2')
sns.set_palette('husl')

# Remove spines
sns.despine()
sns.despine(left=True, bottom=True)

# Reset
sns.reset_defaults()

```

PLOT SELECTION DECISION TREE

Start → What do you want to show?

```

├─ Distribution of ONE variable
│   ├── Numerical
│   │   ├── Basic distribution → histplot() or kdeplot()
│   │   ├── With outliers → boxplot()
│   │   ├── Detailed shape → violinplot()
│   │   └── Percentiles → ecdfplot()
│   └── Categorical
│       └── Frequency → countplot()
├─ Relationship between TWO variables
│   ├── Num × Num
│   │   ├── Correlation → scatterplot()
│   │   ├── With trend → regplot()
│   │   ├── + Marginals → jointplot()
│   │   └── Dense data → hexbin
│   ├── Num × Cat
│   │   ├── Distributions → boxplot() or violinplot()
│   │   ├── All points → stripplot() or swarmplot()
│   │   ├── Aggregated → barplot()
│   │   └── Trend → pointplot()
│   └── Cat × Cat
│       ├── Counts → countplot(hue=...)
│       └── Aggregated → heatmap(pivot)
├─ MULTIPLE variables at once
│   ├── All pairs → pairplot()
│   ├── Correlations → heatmap(df.corr())
│   └── With clustering → clustermap()
└─ Compare across CATEGORIES
    ├── Any plot → FacetGrid()
    └── Categorical → catplot()

```

COMMON PATTERNS

Pattern 1: Explore Distribution

```
# Single variable
sns.histplot(data=df, x='Age', kde=True)

# By category
sns.histplot(data=df, x='Age', hue='Survived', multiple='layer', alpha=0.5)

# Multiple views
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
sns.histplot(data=df, x='Age', ax=axes[0])
sns.boxplot(data=df, y='Age', ax=axes[1])
sns.violinplot(data=df, y='Age', ax=axes[2])
```

Pattern 2: Compare Groups

```
# By one category
sns.boxplot(data=df, x='Pclass', y='Age')

# By two categories
sns.boxplot(data=df, x='Pclass', y='Age', hue='Survived')

# Across multiple categories
sns.catplot(data=df, x='Pclass', y='Age', col='Sex', kind='box')
```

Pattern 3: Relationship Analysis

```
# Basic relationship
sns.scatterplot(data=df, x='Age', y='Fare')

# With third variable
sns.scatterplot(data=df, x='Age', y='Fare', hue='Survived', alpha=0.6)

# With trend
sns.regplot(data=df, x='Age', y='Fare')

# With marginals
sns.jointplot(data=df, x='Age', y='Fare', kind='scatter')
```

Pattern 4: Overview of All Variables

```
# Pairwise relationships
sns.pairplot(data=df, hue='Survived')

# Correlations
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
```

Pattern 5: Complete Categorical Analysis

```
# Distribution
sns.countplot(data=df, x='Pclass')

# Survival rate
```

```
sns.barplot(data=df, x='Pclass', y='Survived')

# By another category
sns.barplot(data=df, x='Pclass', y='Survived', hue='Sex')

# Across categories
sns.catplot(data=df, x='Pclass', y='Survived', col='Embarked', kind='bar')
```

TITANIC-SPECIFIC EXAMPLES

Most Important Plots for Titanic

```
# 1. Age distribution
sns.histplot(data=df, x='Age', bins=30, kde=True)

# 2. Survival by class
sns.countplot(data=df, x='Pclass', hue='Survived')

# 3. Survival rate by class
sns.barplot(data=df, x='Pclass', y='Survived')

# 4. Survival by sex
sns.barplot(data=df, x='Sex', y='Survived')

# 5. Age vs Fare
sns.scatterplot(data=df, x='Age', y='Fare', hue='Survived', alpha=0.6)

# 6. Fare by class
sns.boxplot(data=df, x='Pclass', y='Fare')

# 7. Age by class and sex
sns.violinplot(data=df, x='Pclass', y='Age', hue='Sex')

# 8. All relationships
sns.pairplot(data=df[['Age', 'Fare', 'SibSp', 'Parch', 'Survived']],
             hue='Survived')

# 9. Correlation matrix
sns.heatmap(df.corr(), annot=True, fmt='.2f', cmap='coolwarm')

# 10. Survival by class and sex
sns.catplot(data=df, x='Pclass', y='Survived', hue='Sex', kind='bar')
```

QUICK TIPS

Do's ☐

- Use `histplot()` for distributions (replaces `distplot`)
- Use `regplot()` for trends (simple) or `lmplot()` (with facets)
- Always specify `data=df` parameter
- Use `hue` for third variable coloring
- Use `col` and `row` for faceting

- Add `alpha` for overlapping points
- Use `palette` for custom colors
- Call `plt.tight_layout()` before saving

Don'ts ☐

- Don't use `distplot()` (deprecated, use `histplot`)
- Don't forget to handle missing values
- Don't make pie charts (use bar charts instead)
- Don't use default colors for color-blind audiences
- Don't over-plot (use `alpha`, `jitter`, or aggregation)
- Don't forget axis labels and titles

Performance Tips ☐

- Use `striplot()` not `swarmplot()` for large data (>1000 points)
- Use `hexbin` for very large scatter data (>10000 points)
- Sample data if needed: `df.sample(1000)`
- Use `ci=None` to skip confidence interval calculation

SAVING FIGURES

```
# Save current figure
plt.savefig('plot.png', dpi=300, bbox_inches='tight')

# Different formats
plt.savefig('plot.pdf') # Vector
plt.savefig('plot.svg') # Vector
plt.savefig('plot.jpg') # Compressed

# Save FacetGrid or other Seaborn objects
g = sns.pairplot(data=df, hue='Survived')
g.savefig('pairplot.png', dpi=300)

# Close figure to free memory
plt.close()
```

End of Cheat Sheet 📄

Practice with Titanic dataset to master these visualizations!